Ch-03

5dB km-1

Attenuation: refers to the reduction ~~of~~ in power of the light signal as it travels through the fiber. It also called transmission looses. It determines maximum transmission distances.

input optical power $= P_i$

output " " $= P_o$

∴ Number of decibels (dB) $= 10 \log_{10} \frac{P_i}{P_o}$

- Power & roots reduces to multiplication & division.
- Addition & subtraction require a conversion to numerical values which may be obtained, → numerical value $= \frac{P_i}{P_o} = 10^{(dB/10)}$

→ ~~$\alpha dB L$~~ $\boxed{\alpha_{dB} L = 10 \log \frac{P_i}{P_o} \ dB\,km^{-1}}$

where, α_{dB} is the signal attenuation per unit.

Material absorption: material absorption is a loss mechanism related to the material composition & fabrication process for the fiber which results in the dissipation of some of the transmitted optical power as heat in the wave webguide.

This reduces the transmitted optical power.

- Type of absorption-

1) Intrinsic → caused by the interaction with one or more of the majore components of the glass. (caused by the natural properties of the fiber material → usually silica glass).

Two main region →

- UV absorption (short wavelength)
- IR absorption (long wavelength.

→ OH (hydroxyl ions) from water dissolve in the glass.

2) Extrinsic Absorption → (caused by the impurities within the glass) → (impurities in the fiber manufacturing).

– (fabrication process এর time-এ হয়).

common impurities – Metal ion (Fe, Cu). (OH^-).

→ * Explain ~~their~~ in details the intrinsic absorption loss mechanism with necessary illustration → def :- caused by the interaction with one or more major components of the glass – called intrinsic absorption.

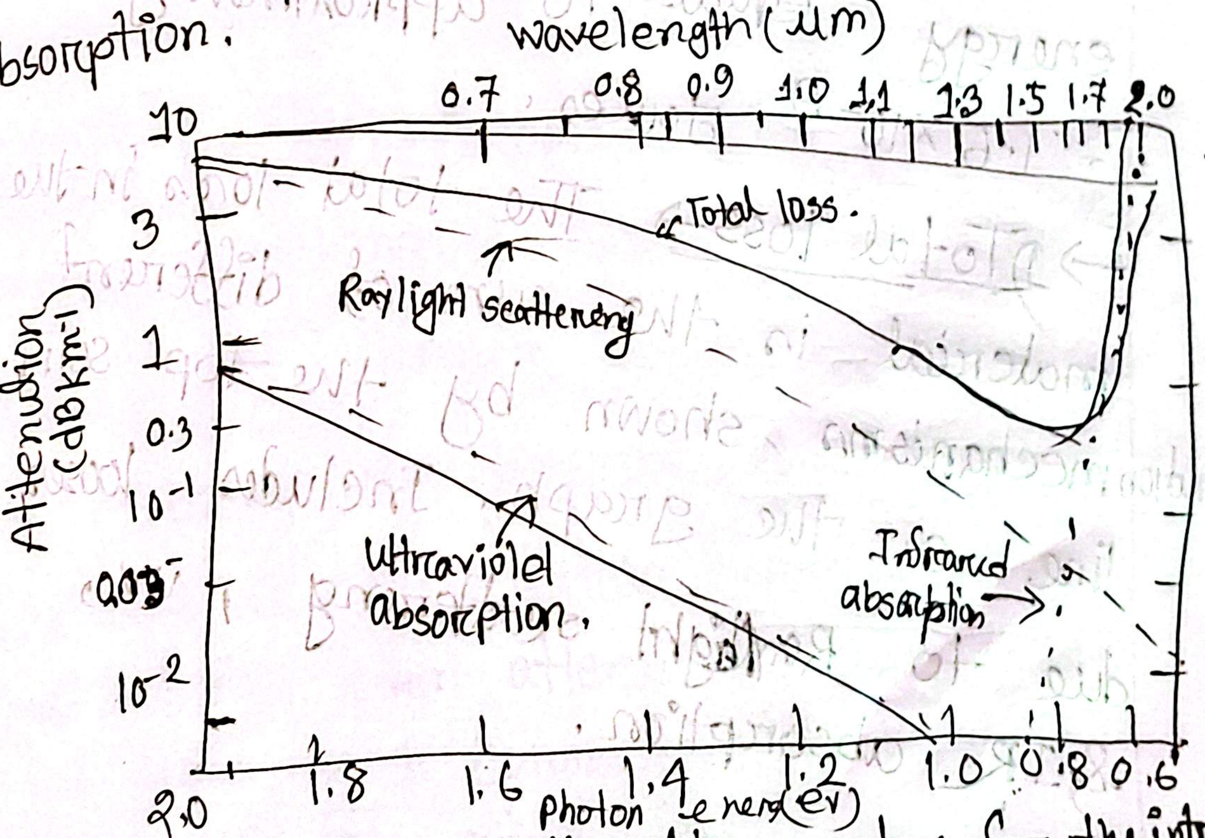


Fig: The attenuation spectra for the intrinsic loss mechanism in pure $GeO_2 - SiO_2$ glass.

→ The graph shows the attenution spectra for the intrinsic loss mechasim. in pure $GeO_2 - SiO_2$ glass. Which is commonly used in optical fibers. The attenuition is measured in dB/km and plotted against both photon energy (eV) & wavelength (μm)

→ The electronvolt is a unit of energy equal to approximately $= 1.61 \times 10^{-19}$ Joules.

→ 1) Total loss → The total loss in the material is the sum of different attenution mechanisms, shown by the top solid line on the graph. Includes looses due to Rayleight scattering, UV, & IR absorption.

2) Rayleight Scattering → curve causes attenuation due to imperfaction & variations in the glass structure.
- The Rayleight scattering curve decreases as the wavelength increases, showing that the effect is more prominent at shorter wavelengths.

3) UV $\xrightarrow{\text{absorption}}$ is an electromagnetic radiation with a wavelength from 10 nm to 400 nm, shorter than that of visible light but ~~longrer~~ longer than X-Rays.

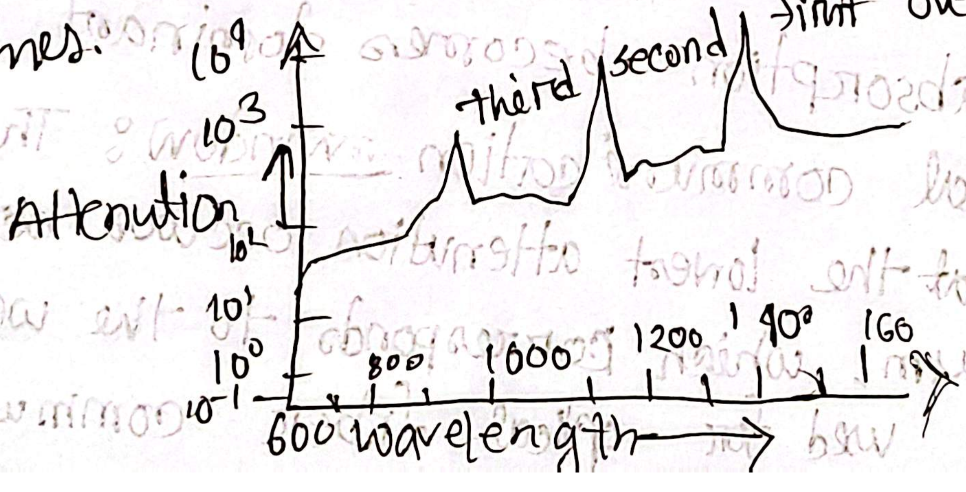
4) IR: is electromagnetic radiation (EMR). longer wavelengths than those of visible light. At longer wavelength (beyond 1.6 um) IR absorption becomes dominant.

5) optical communication window: The graph shows that the lowest attenution occurs between 0.8 um & 1.6 um, which corresponds to the wavelength range used for optical fiber communication.

This is because both Rayleigh scattering & Infrared absorption are minimized in this range, leading to optimal performance.

* Absorption ~~losses~~ losses caused by some of the more common metalic ion impurities, namely chromium (Cr) & (Cu) in the worst valence state can cause attenuation in excess of 1 dB/km in the infrared region.

* Extrinsic - (OH^-) - Another major extrinsic loss mechanism is caused by absorption due to water OH^- dissolved in glasses.



// Absorption losses caused by ~~th~~ some of the more common metalic ion impurities ~~of~~, in glasses, together with the absorption peak wavelength. — namely Cr & Cu in worst valence state can cause attenution in excess of 1 dB km^{-1} in the near infrared region.

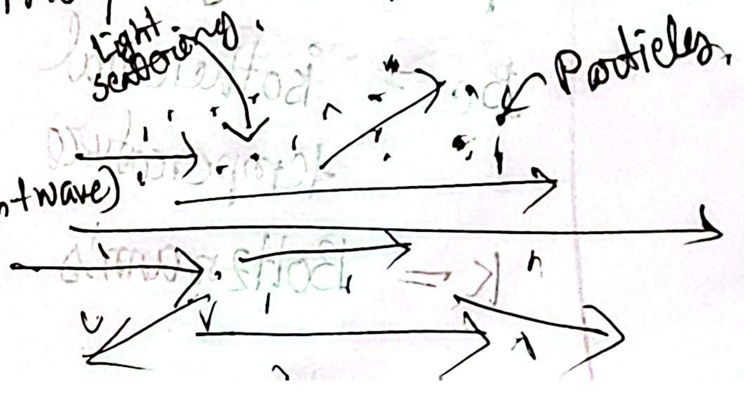
// Linear scattering losses → Linear scattering mechanism are process in which some or all of the optical power from one propageting mode is transfared to another mode in a linear manner.

where optical power is transfered from one propageting mode to another is a loss mechanism

→ The transfer of power often leads to attenution because the light may be coupled into —

① Leaky modes:

② Radiation modes. (Light wave)



→ These mode does not continue to propagated within the fiber core. but is radiated from the fiber.

<u>Reyleigh scattering</u> - is the elastic scattering of light or other electromagnetic radiation. by partices much smaller than wavelength of the radiation.

$$\gamma_R = \frac{8\pi^3}{3\lambda^4} n^8 p^2 \beta_c k T_F.$$

γ_R = Rayleight scattering coefficient.

λ = optical wavelength.

n = refractive index of medium.

p = average photoelastic coefficient

β_c = isothermal compressibility at affective temperature T_F.

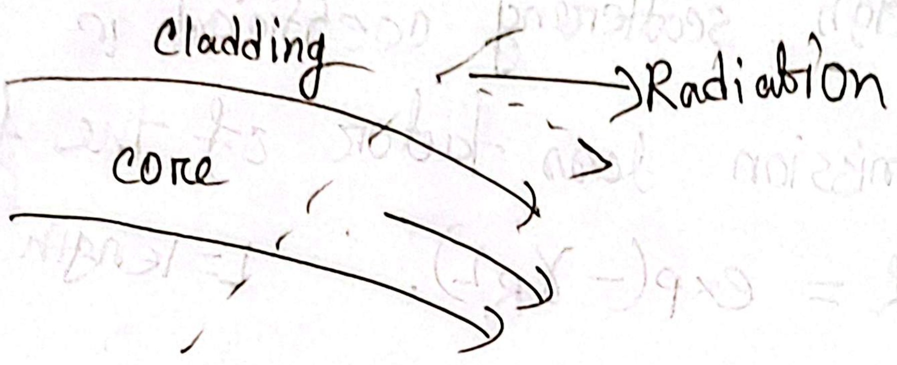
K = Boltzmann's constant.

CamScanner

Rayleigh scattering coefficient is related to transmission loss factor of the fiber $\mathcal{L}$.

$$\mathcal{L} = \exp(-\gamma_R L).$$ L = length of the fiber.

Fiber bend loss → Optical fibers experience radiation loss when they are bent or curved along their path. This happens because part of the light energy, present in the evanscent field, extends into the cladding at the bend exceeding the velocity of light in the cladding and hence the guidance mechanism is inhibited, which causes light energy to be radiated from the fiber.



→ ~~Different mode~~ fiber.

→ Radiation attenution coefficient

$\alpha_r = C_1 \exp(-C_2 R)$

R = radius of curvature of fiber bend

C_1, C_2 = constant in multimod fibers at a critical radius of curvature R_c

$$R_c \simeq \frac{3 n_1^2 \lambda}{4\pi (n_1^2 - n_2^2)^{3/2}}$$

// Simulated Brillion scattering :-

$P_B = 4.4 \times 10^{-3} d^2 \lambda^2 \alpha_{dB} V$ watts.

// Raman scattering - $P_R = 5.9 \times 10^{-2} d^2 \lambda \alpha_{dB}$ W

What is an optical attenution used for -

1. Evaluting transmission distance : Optical attenution helps determine the maximum distance an optical signal can travel through.

2. Compare with metalic conductor :- It allows for comparing fiber optics with metalic cable. Lower attenution in fiber make them more suitable for long distance communication.

3. Signal quality assesment : Attenution measure the reduction in signal strength, which helps access and maintain quality.

4. System design & optimizations : Knowing attenution of fiber optic we can ensure they are optimized for distance.

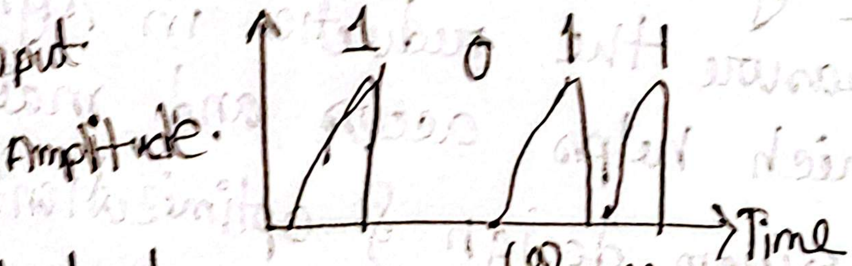
5. Loss mechanism Analysis : Understanding attenution helps to address various loss mechanism.

the reduction in signal stre

6. Performance Monitoring → Regular attenuation measurement help monitor & maintain the performance of the fiber optic network.

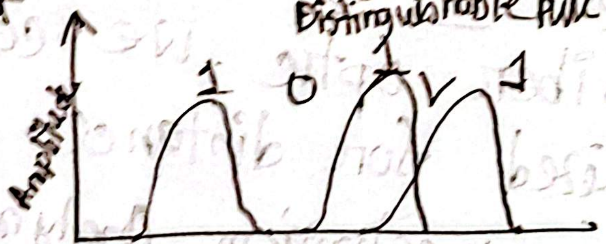
Dispersion: refers to the spreading of light pulse as they travel down the fiber. It's causes distortion for both digital & analog transmission.

a) fiber input



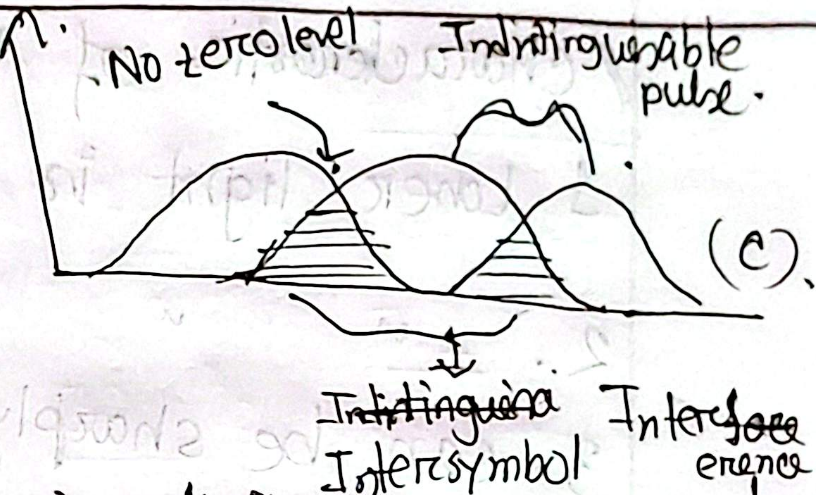
(a) Distinguishable pulse.

b) fiber output at a distance L1



(b)

(c) fiber output at a distance L2 > L1



- an illustration using digital bit 1011 of broadening of light.

(a) fiber input (b) fiber output at a distance L, c) fiber output at distance L2>L1

// The energy difference due to the transition of electron leads two process.

1. Radiative process : In this process photons are emitted.

2. Non radiative process; Process lead to the creating of heat.

// Characteristic of Laser -

1. Laser light is highly coherent

2. " " " " monochromatic.

3. can be sharply focused.

4. Brightness.